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RESEARCH-ARTICLE

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LoRa is widely adopted for IoT connectivity. While integrating sensing and communication (ISAC) in LoRa is crucial for future development, existing approaches fail to implement ISAC with ambient LoRa traffic due to low data rates. Relying solely on the limited preamble for sensing proves insufficient, as it often requires dedicated sensing devices or protocol modifications. To address these challenges, we theoretically analyze the trade-off between communication and sensing ambient LoRa traffic. By introducing the Sensing-Communication Ratio (SCR) and per-symbol sensing entropy, we provide new metrics to quantify sensing capacity. Building on these insights, we propose SenLoRa, the first practical LoRa ISAC system that integrates seamlessly with existing communication infrastructure, enabling a range of sensing applications with optimized SCR. The core of SenLoRa lies in confidence symbol extraction, which uses a likelihood probability ratio to identify high-confidence symbols from payload for both communication and sensing entropy enhancement. We also design an incentive strategy to obtain additional sensing entropy. Real-world case studies demonstrate that SenLoRa achieves a respiration detection error as low as 0.2 bpm and over 90% accuracy in walking detection, while maintaining standard LoRa communication. These results validate SenLoRa as a practical and efficient ISAC solution for LoRa-based IoT systems.

CCS Concepts: • **Human-centered computing → Ubiquitous and mobile computing systems and tools**.

Additional Key Words and Phrases: LoRa, Integrated Sensing and Communication (ISAC), Sensing, Soft Demodulation, Log Likelihood Ratio (LLR)

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1 INTRODUCTION

LoRa, as a low-power wide area network technology, has become indispensable in IoT, connecting billions of devices[20]. Its excellent performance has led to widespread adoption in various fields such as cold food

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storage[3, 26], cold chain management (CCM)[16, 40], cultural relics storage[14], and chemical plant monitoring[1]. These applications often involve high-value items stored in specialized areas with strict environmental controls, requiring more frequent reporting of conditions.

The field of RF-based sensing has been thoroughly explored by researchers[10, 24, 25, 56] and LoRa-based wireless monitoring technology has become fully integrated into smart logistics infrastructures as a core component, particularly in warehouse environments. Existing research confirms its effectiveness in ensuring the quality and stability of perishable goods through continuous monitoring of key environmental parameters such as temperature, humidity, and gas concentrations[12, 15, 26, 41]. Notable applications include Jedermann et al.'s[26] implementation of a 4-node network for minute-by-minute dynamic airflow analysis (1 reading/minute) in cold storage, and Thermokon's[41] 30-node temperature and humidity monitoring system, which further validated the technology's reliability in large-scale warehouse environments. However, from a safety management perspective, current cold chain management presents two significant challenges: First, ensuring cargo security often relies on lone workers performing night patrols in cold storage facilities, exposing these individuals to potential risks inherent in isolated work environments. Second, while traditional video surveillance systems[43] can supplement oversight, the low-temperature, high-humidity conditions require specialized cameras[37], substantially increasing implementation costs. Consequently, there is a pressing need to develop a low-cost, passive monitoring solution for the walking and vital signs (respiration) for cold storage workers by leveraging the existing LoRa infrastructure, thus enhancing workplace safety standards.

Fortunately, the Integrated Sensing and Communication (ISAC) paradigm offers a compelling opportunity to imbue communication systems with sensing. Although much current ISAC research focuses primarily on WiFi[19, 22, 28], prompting initial consideration of these systems for our target scenario, existing WiFi-based ISAC approaches are generally unsuitable for the demands of perishable goods warehousing. This unsuitability arises primarily from two key limitations: 1) Insufficient Range: The typical sensing range of WiFi ISAC systems, often on the order of 10 meters, falls significantly short of providing adequate coverage across the expansive areas characteristic of standard warehouses, which can easily exceed (e.g., 200-1000 m^2). 2) Signal Interference: The complex internal environment of warehouses, densely packed with shelving, diverse goods (potentially with high water content or metallic packaging), and structural elements, introduces substantial signal obstruction and multipath interference, which can severely degrade the reliability and accuracy of WiFi-based sensing.

However, existing LoRa-based ISAC approaches also have limitations. They either require the use of pre-known traffic to extract sensing information or require additional hardware to eliminate hardware effects in the received signals. For example, Song et al.[38] used dual-antennas and helper nodes for transmitter tracking, while Huang et al.[23] embedded LoRa signals in FMCW radar waveforms. Therefore, these approaches cannot utilize ambient LoRa traffic with commercially deployed nodes in real-world scenarios.Neither of these approaches is applicable in the current context. The first requires adding helper nodes and antennas, effectively doubling the cost at minimum. The second, while offering adequate sensing precision, suffers from a severely limited operational range of just 20 meters.

	WiFi[19]	**FMCW radar[23]**	**LoRa**[38]
More Hardware	No	No	Yes
Power Consumption	High	High	Low
Sensing Range	Small-Scale Indoor	Typical Range (10-30m)	150m
Communication Range	Typical Range (<100m)	16m	Typical Range (2-5km)

Table 1. Comparison of different ISAC System

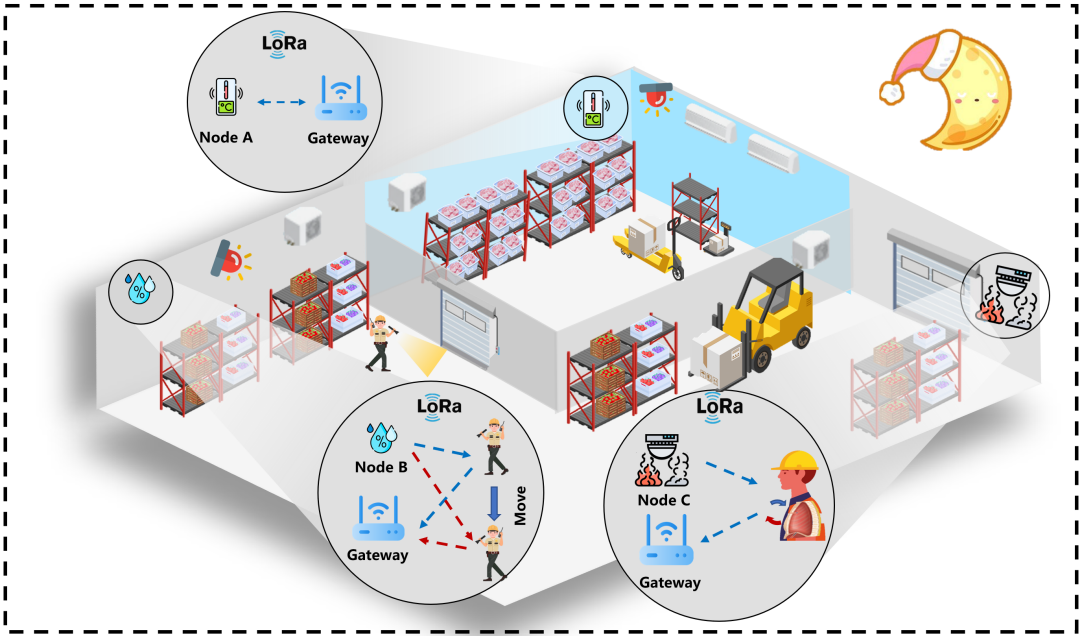


Fig. 1. Passive sensing concept for lone worker safety monitoring in cold storage using ambient LoRaWAN signals. Existing nodes (like Node A for environment) communicate with the gateway. Our system analyzes these signals to detect worker walking based on movement disruptions (Node B example) and respiration based on subtle signal changes from breathing (Node C example), repurposing communication for safety.

To overcome this limitation, our goal is to develop a seamless ISAC system based on commercial LoRa infrastructure. As illustrated in Fig. 1, this concept is depicted within a typical smart warehouse environment. Here, multiple COTS LoRa nodes have already been deployed. These nodes periodically communicate environmental data or alerts to a central LoRa gateway according to LoRaWAN. Our proposed system, SenLoRa, operates non-intrusively within this existing setup. By passively sniffing and analyzing the ambient LoRa signals transmitted between these unmodified nodes and the gateway, SenLoRa extracts valuable secondary sensing information – such as walking and dynamics of personnel or the respiration – without requiring any hardware modifications, protocol changes, or additional dedicated sensing devices. This approach effectively reuses the existing communication infrastructure for enhanced situational awareness and operational intelligence.

Challenges & Solutions: A fundamental challenge in developing ISAC systems lies in the inherently conflicting requirements between wireless communication and sensing regarding multipath components[33]. In communication systems employing hard-decision demodulation, the demodulator typically treats the main path signal as the desired signal, while considering multipath components as interference. When multipath signals have similar strength to the main path signal, they generate adjacent peaks in the frequency domain. In this case, demodulators find it difficult to identify the true modulation symbol, which increases the probability of demodulation errors[11, 39]. Unlike communication systems, sensing systems aim to extract multipath components for sensing information. To achieve this, they typically use extra hardware like dual-antennas[53] or dual receivers[38] to handle noise and capture useful multipath signals. This apparent contradiction raises a critical question: How can we balance these competing requirements within a single system?

To address this challenge, our simulation experiments reveal that higher Signal-to-Sensing-Noise Ratio (SSNR) does not necessarily translate to better sensing capability in certain scenarios. Based on this observation, we

first establish a theoretical framework to analyze the trade-off between communication reliability and sensing performance. We propose the Sensing-Communication Ratio (SCR), which quantifies the relationship between the target reflection power used for sensing and the overall received signal power. This lays the foundation for refining our sensing theory.

Another challenge is that sensing information primarily relies on accurately extracting the differences between received signals and known reference signals. The most straightforward approach is to utilize the preamble sequence of LoRa packets, which consists of standard up-chirps that serve as ideal reference signals. However, the standard protocol only includes 8 chirps in the preamble, which is insufficient for high-quality sensing requirements.

To address this challenge, inspired by information theory, we recognize that the sensing entropy carried by the preamble sequence alone is insufficient for detecting complete activity cycles. To fully utilize the sensing information within the entire packet, we propose a soft-decision demodulation scheme based on Log Likelihood Ratios (LLR). This scheme maps FFT results to probability intervals, not only enhancing demodulation accuracy but also pioneering a new approach to extract sensing information from the payload portion.

Another key challenge lies in maintaining the temporal continuity of the sensing information. Although we can extract complete sensing information from individual packets, the temporal resolution of single-packet data remains insufficient for specific tasks that require continuous observation, such as breathing detection. This limitation arises because physiological activities like human breathing exhibit long-period characteristics that demand observation and analysis over extended time scales.

To overcome this challenge, we propose an adaptive triggering strategy based on sensing entropy distribution. This strategy introduces a concentration parameter κ as the core regulation metric, which dynamically adjusts the packet transmission times of end devices by evaluating real-time entropy distribution characteristics. This mechanism not only adapts packet quantity according to different sensing task requirements to ensure sensing accuracy, but also maintains additional energy consumption within acceptable limits through precise parameter control, effectively balancing sensing performance with device battery life.

In summary, SenLoRa addresses these challenges by extracting confidence symbols from unknown payloads using a likelihood probability ratio, which serves a dual purpose: enhancing decoding capacity during the communication phase and increasing sensing entropy during the sensing phase. Furthermore, when normal traffic lacks sufficient sensing entropy, SenLoRa actively injects probing packets to augment sensing capabilities while maintaining communication performance, achieving seamless integration of sensing and communication. In summary, our contributions are as follows:

(1) We propose SenLoRa, the first practical LoRa ISAC system that can be seamlessly integrated into existing communication systems to enable various sensing applications. By leveraging unknown payloads, it enhances decoding capacity during communication and increases sensing entropy during sensing without modifying LoRaWAN protocols or requiring additional hardware..
(2) We theoretically analyze the trade-off between communication and sensing, introducing the Sensing-Communication Ratio (SCR) and per-symbol sensing entropy to quantify sensing capacity during ambient LoRa communication. To enhance sensing entropy while maintaining communication capacity, we propose confidence symbol extraction and packet incentive strategy to ensure reliable data transmission alongside accurate sensing.
(3) We develop a prototype of SenLoRa and conduct real-world experiments. The results show that SenLoRa significantly enhances LoRa sensing entropy by extracting a substantial number of confidence symbols and improves communication performance by 4x compared to standard demodulation in LoRaWAN. Case studies in real-world sensing applications demonstrate that SenLoRa achieves a respiration detection error

of 0.2 breaths per minute (bpm) and over 90% accuracy in walking detection, without compromising normal LoRa communication.

2 UNDERSTANDING LORA SIGNAL FOR COMMUNICATION AND SENSING

LoRa Demodulation: The receiver detects the preamble of the LoRa frame by correlation on a continuous base up-chirps[49]. After successful detection, the frames will be aligned through SFD and then demodulation of the Payload. A chirp arriving at the receiver through the wireless channel can be expressed as

$$Rx(t) = H(t)Tx(t) + n(t), t \in [0, T), \tag{1}$$

where the $H(t)$ is the channel effect, and the $n(t)$ is the noise. The standard LoRa demodulator realizes *de-chirp* operation by multiplying the received chirp with base down-chirp $C^{-1}(t)$, which is the conjugate of the base up-chirp, converting it into a single tone signal:

$$D(t) = Rx(t) \cdot C^{-1}(t) \tag{2}$$

Then it is converted to the frequency domain through FFT (Fast Fourier Transform),

$$Z[f] = FFT(D(t)), f = [0, 2^{SF} - 1]. \tag{3}$$

and find the index of the bin with the maximum peak, which is mapped to a symbol $\hat{s}$.

$$\hat{s} = \arg\max_{f} |Z[f]|) \tag{4}$$

2.1 LoRa Communication

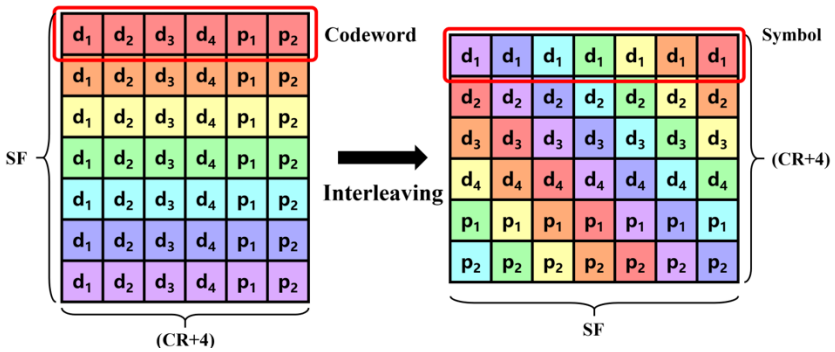


Fig. 2. The LoRa interleaving block formation process when CR=2.

LoRa Decode: Fig. 2 shows the encoding and decoding process of LoRa. For data given by the upper layer protocol, LoRa PHY will first divide it into a sequence of nibbles $[d_1, d_2, ..., d_n]$, which is half of a Byte (i.e. 4 bit), and add CR Hamming check bits behind d_i based on the encoding rate to form a codeword of $4 + CR$ bits. Then select consecutive SF codewords forming $SF \times (4 + CR)$ bit matrix for diagonal interleaving like the shifting form of the shading grid and obtain $(4 + CR)$ LoRa symbols of SF bit. Finally, through gray demap obtain the modulation transmission sequence $[s_1, s_2, ..., s_m]$.

2.2 LoRa Sensing

In LoRa sensing modeling[53], the received signal $Rx(t) = e^{j(\pi kt^2+j2\pi f_c t)} \sum_{n=1}^{N} a_n(t)e^{-j2\pi f_c \tau_n(t)} + n(t)$, which describes the signal propagation along multiple paths with corresponding delays, and the superposition of signals at the receiver. By multiplying the received signal with $e^{-j2\pi f_c' t}$, we obtain the sensing signal as follows:

$$Rx'(t) = e^{j(\pi kt^2+\theta_c+\theta_s)} \sum_{n=1}^{N} a_n(t)e^{-j2\pi f_c \tau_n(t)} + n(t) \tag{5}$$

where θ_c and θ_s are the phase errors introduced by the carrier frequency offset (CFO) and sampling frequency offset (SFO), respectively. Mathematically, the latter part $\sum_{n=1}^{N} a_n(t)e^{-j2\pi f_c \tau_n(t)}$ represents the signal envelope which is related to the target's movement. The multipaths components include both dynamic vector $H_d = a(t)e^{-j2\pi f_c \tau(t)}$ and static vectors $H_s = \sum_{i\in P_s} a_i e^{j2\pi f \tau_i(t)}$.

3 QUANTIFYING SEN-COM CAPABILITY

In this section, we first systematically analyze the contradiction between communication and sensing in ISAC systems. To quantify the balance between sensing and communication, we propose the Sensing-Communication Ratio (SCR).

3.1 Contradiction between Communication and Sensing

Wireless Communication[33] highlights that typical wireless systems experience BER of 10^{-2} under standard SNR conditions, primarily due to multipaths propagation. Despite LoRa's inherent noise resistance capabilities, its communication performance can still be adversely affected under severe multipaths conditions induced by environmental factors or motion[11, 39]. Because the receiver cannot distinguish these multipaths, this leads to the generation of interference. Consequently, mitigating multipath-induced communication impairments is crucial in wireless system design.

Consider a received signal $Rx(t)$, where $\sum_{n=1}^{N} a_n(t)e^{-j2\pi f_c \tau_n(t)} + n(t)$ are the component encapsulates multipaths effects and noise. Here, $n(t)$ is characterized by a Gaussian distribution with zero mean and variance σ^2. To mitigate interference during the demodulation process, it is imperative to minimize the impact of this component through design of modulation, demodulation and coding schemes. Consequently, the communication system must address the following optimization problem: $\min_{a_n(t),\tau_n(t)} \mathbb{E}\left[\left|\sum_{n=1}^{N} a_n(t)e^{-j2\pi f_c \tau_n(t)} + n(t)\right|^2\right]$ s.t. $\sum_{n=1}^{N} |a_n(t)|^2 \le P$. This can be reduced to the following simplified form:

$$\min_{a_n(t),\tau_n(t)} \sum_{n=1}^{N} \mathbb{E}[|a_n(t)|^2] \quad \text{s.t.} \sum_{n=1}^{N} |a_n(t)|^2 \le P \tag{6}$$

However, from the perspective of the sensing system[52], the multipaths $\sum_{n=1}^{N} a_n(t)e^{-j2\pi f_c \tau_n(t)}$ component in Eq. 5, represents the signal envelope related to target motion. This includes a static vector $H_s = \sum_{i\in P_s} a_i e^{j2\pi f \tau_i(t)}$ and a dynamic vector $H_d = a(t)e^{-j2\pi f_c \tau(t)}, \tau(t) = \frac{d(t)}{c}$ .The dynamic vector H_d contains motion information of the sensed object. Moreover, SSNR[52] has been proven to be an effective indicator for quantifying sensing capability: $SSNR = \frac{\mathbb{E}(|a(t)|^2)}{\sigma^2}$.

The goal of sensing system is to achieve the highest possible SSNR, i.e., to solve the following optimization problem:

$$\max_{a_n(t),\tau_n(t)} \mathbb{E}[|a(t)|^2] \quad \text{s.t.} \sum_{n=1}^{N} |a_n(t)|^2 \le P \tag{7}$$

The optimization objectives of communication and sensing systems exhibit fundamental differences and potential conflicts. While communication systems strive to minimize channel-induced uncertainties by reducing multipaths effects and noise impacts, sensing systems aim to maximize the power of dynamic multipath components for improved target detection and tracking. Therefore, the design of Integrated Sensing and Communication (ISAC) systems necessitates a meticulous balance between these conflicting objectives.

3.2 Sensing-Communication Ratio (SCR)

Definition. Based on the analysis in Sec. 3.1, we need to identify an appropriate balance between sensing and communication. Therefore, we propose a metric called the Sensing-Communication Ratio (SCR) to quantify this balance. It is defined as the ratio of the power of target reflection to the received signal, divided by the noise power:

$$SCR = \gamma = \frac{\mathbb{E}\left(|H_d/(H_s + H_d)|^2\right)}{\mathbb{E}\left(|\boldsymbol{n}(\boldsymbol{t})|^2\right)} \tag{8}$$

The equation demonstrates that a higher ratio of target reflection power to received signal power yields a larger SCR, indicative of enhanced sensing capability. Given the wide dynamic range of wireless signal strengths, SCR is typically expressed in decibels (dB) as γdB = 10 log10 γ.

While the SCR formula is straightforward, its importance should not be understated. It provides a vital complement to the integrated sensing and communication theory for LoRa systems. Notably, despite extensive research in LoRa sensing and communication in recent years, there has been a paucity of formally defined metrics quantifying the balance between sensing and communication functionalities. The SCR addresses this gap in the literature, offering a standardized measure for evaluating the sensing-communication trade-off in integrated systems.

Table 2. Simulation Parameters

	$\lvert H_s\rvert$	$\lvert H_d\rvert$	σ
Case 1	100	0.1-10	1
Case 2	1-100	1	1

Verification. In this part, we validate the SCR as an appropriate metric for assessing ISAC capability through simulations. We generate signal samples simulating human respiration and use SNR and SSNR as benchmarks for communication and sensing performance, respectively[52]. In our simulations, we carefully design two combinations of ($|H_s|$) and ($|H_d|$), as detailed in Table 2. Using Eq. 8, we calculate the SSNR, SNR, and SCR for each scenario, and the simulation results are illustrated in Fig. 3. In case 1, an increase in SCR corresponds to an improvement in SSNR, indicating enhanced sensing capability with stable communication performance. In case 2, an increase in SCR is accompanied by a decrease in SNR, signifying reduced communication capability while sensing performance remains unchanged. Thus, we conclude that the SCR is closely related to both communication and sensing performance and is a suitable metric for quantifying the balance between communication and sensing in LoRa systems.

3.3 Sensing Entropy

Through the above analysis, we have confirmed that SCR is an effective metric for evaluating sensing-communication capability. Meanwhile, SSNR serves as a valid indicator of sensing capability[52], reflecting the sensing quality at a given moment. However, for human activities (such as respiration) that induce periodic Signal envelope

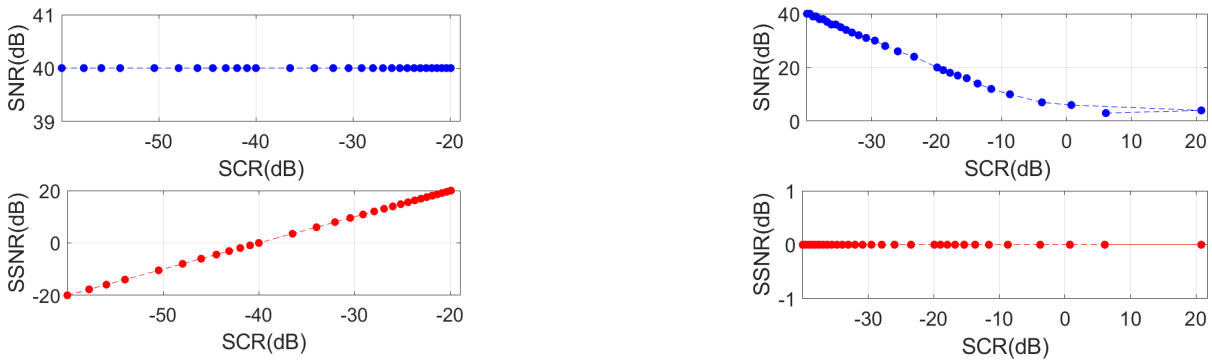
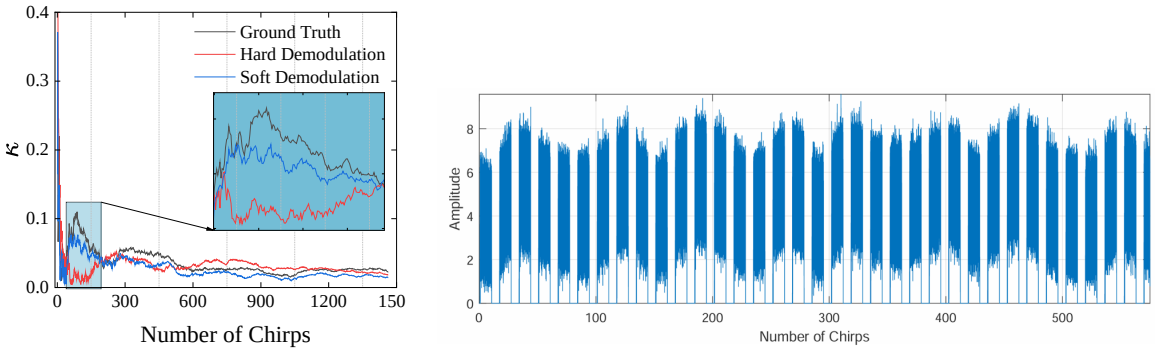


Fig. 3. Simulation results. (**Left**) Case 1 plots showing stable SNR (≈40dB) with increasing SSNR. (**Right**) Case 2 plots showing dropping SNR with constant SSNR. Interpretation: Case 1 indicates improved sensing without compromising communication, while Case 2 suggests communication degradation with limited sensing benefits.



(a) Relationship between Chirps and κ

(b) Relationship between Chirps and CSI

Fig. 4. Sensing Entropy Convergence vs. Chirp Count and Corresponding CSI Amplitude Envelope

drift in LoRa signals, even when SSNR is high at a particular time, the task may fail due to insufficient temporal resolution caused by symbol duration constraints. To assess whether the information carried by a signal segment is sufficient to complete the sensing task, we introduce the concept of sensing entropy.

Definition. To quantify the regularity and predictability inherent in the time series derived from the sensing signal (e.g. fluctuations in received signal strength or phase), we can utilize Sample Entropy[36]. Sample Entropy assesses the complexity of a time series by measuring the negative natural logarithm of the conditional probability that two sequences similar for m points remain similar for $m+1$ points, within a tolerance r, without allowing self-matches. Conceptually, it is calculated as follows:

$$SE(H_d(t), m, r) = \kappa = -\ln\left(\frac{A^{m+1}(r)}{B^m(r)}\right) \tag{9}$$

Where m is the embedding dimension and r is the tolerance. $B^m(r)$ counts the number of similar pattern pairs of length m, and $A^{m+1}(r)$ counts the number of similar pattern pairs of length $m+1$. The detailed derivation process can be found in the Appendix 12.

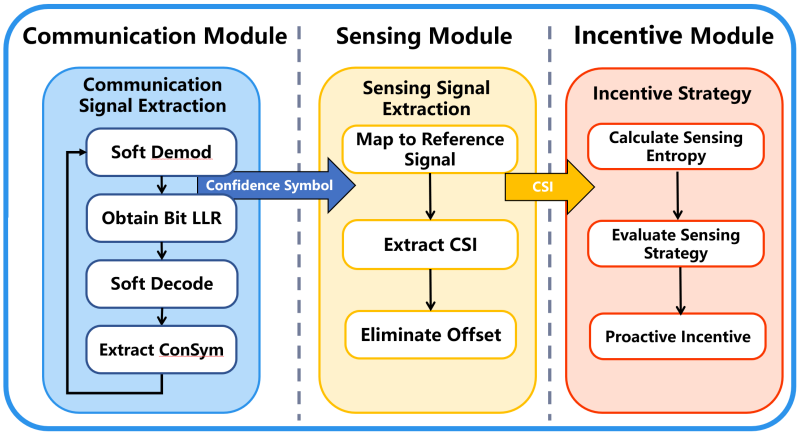


Fig. 5. Architecture of SenLoRa, sniffing uplink data packets from COTS nodes and sensing human activities.

Verification. Fig. 4 plots the variation in sensing entropy κ against the number of received chirps. The figure compares three curves: Ground Truth (black), Hard Demodulation (red), and Soft Demodulation (blue). Initially, when the number of chirps is small (see inset), the entropy values for Ground Truth and Soft Demodulation are relatively high and fluctuate significantly, indicating that the collected data is insufficient to reliably represent the underlying periodic activity. The Hard Demodulation curve, in contrast, starts from lower values. As the cumulative number of chirps increases, all three curves begin to converge. Notably, after approximately 600 chirps, the entropy value κ stabilizes, especially for the Soft Demodulation curve which closely tracks the Ground Truth. This suggests that sufficient information has been captured by this point to sense the periodic activity.Fig. 4(b) displays the corresponding CSI amplitude envelope for the first approximately 600 chirps. The plot clearly shows multiple signal bursts (representing chirps within transmitted LoRa packets) separated by low-amplitude gaps (the intervals between packet transmissions). Crucially, the peak amplitude of these signal bursts exhibits periodic fluctuations. By observing this periodic fluctuation across multiple packets, the frequency of the underlying activity (e.g. breathing rate) can be determined. The convergence phenomenon observed around 600 chirps in Fig. 4(a) implies that collecting data spanning several packets visible in Fig. 4(b) is necessary to obtain a stable entropy metric. This insight, combined with the fact that there are intervals between packets, indicates that by optimizing the triggering strategy, it is possible to sense respiratory activity using the minimum number of packets required to reach entropy convergence, thereby saving energy.

4 SYSTEM OVERVIEW

As shown in Fig. 5, SenLoRa can sense human activities by sniffing uplink data packets from COTS nodes. The system architecture of SenLoRa mainly includes three modules: communication, sensing and incentive. This design allows SenLoRa to simultaneously achieve communication and sensing functions without requiring additional hardware or modifying the LoRaWAN protocol.

Specifically, in communication module, SenLoRa first extracts the communication signal through soft demodulation methods. This process includes obtaining the Log Likelihood Ratio (LLR) of bits, which serves as a key indicator for assessing bit confidence. Subsequently, the system performs soft decoding operations and extracts Confidence Symbol (ConSym). The purpose of these steps is to optimize LoRa communication performance and provide reliable reference signals for the subsequent sensing stage.

In sensing module, the system first maps the extracted signal to a reference signal, then eliminates various offsets, such as Carrier Frequency Offset (CFO) and Sampling Time Offset (STO). This step helps eliminate the modulation frequency in the payload, thereby obtaining more accurate Channel State Information. Finally, the system extracts the channel state information, laying the foundation for subsequent sensing tasks.

In incentive module, SenLoRa calculates the sensing entropy carried in the current uplink data packet to evaluate sensing quality. If the sensing entropy is insufficient, SenLoRa will utilize beacons in LoRaWAN to request nodes retransmit sufficient uplink frames.

This three-stage design enables SenLoRa to achieve seamless integration of communication and sensing on the existing LoRa network architecture. By fully utilizing the information in communication packets, SenLoRa not only guarantees communication functionality but also achieves continuous environmental sensing, filling the gap in LoRa ISAC.

5 COMMUNICATION MODULE

In this section, we optimize communication performance through enhancing LoRa's demodulation and decoding capabilities by computing LLRs and extracting pilots for channel sensing. The receiver mitigates factors leading to symbol misinterpretation, but hardware offset, noise, and side lobe leakage can result in multiple similar-height peaks in the spectrum. LoRa's hard decision demodulation, relying solely on the highest peak, can lead to significant error rates. Despite using Gray code mapping to reduce misclassification between adjacent bins, this approach is insufficient for correct decoding in adverse channel conditions. To address this, we propose soft demodulation to comprehensively consider the energy distribution across frequency domain bins, and iteratively select the optimal confidence symbol to greatly improve communication capability.

5.1 LLR Calculation and Soft Demod

In other wireless networks that use M-QAM modulation, such as WiFi, 3G/4G/5G[42, 55], the LLR of the corresponding bit under the current received symbol can be calculated based on probability during demodulation. For a transmitted symbol reaching the receiver through the wireless channel, the received symbol $\mathbf{r}$ is obtained, and the LLR of the corresponding k-th bit LLR_k is calculated as follows:

$$LLR_k(\mathbf{r}) = \log \frac{P(b_k = 1 \mid \mathbf{r})}{P(b_k = 0 \mid \mathbf{r})} = \log \frac{P(\mathbf{r} \mid b_k = 1)P(b_k = 1)/P(\mathbf{r})}{P(\mathbf{r} \mid b_k = 0)P(b_k = 0)/P(\mathbf{r})} = \log \frac{\sum_{s \in S_{b_k=1}} P(\mathbf{r} \mid s)}{\sum_{s \in S_{b_k=0}} P(\mathbf{r} \mid s)} \tag{10}$$

where $k = 0, 1, ..., log_2 M$, and $S_{b_k=1}$ represents the symbol subset whose k-th bit is 1 and similarly $S_{b_k=0}$ represents the subset of rest symbols whose k-th bit is 0. Both of them have a $M/2$ number of symbols.

In the above formula Eq. 10, the posterior probability $P(b_k|\mathbf{r})$ is obtained through Bayes' rule to obtain the prior probability $P(\mathbf{r}|b_k)$. Normally, sending different data is equally likely, that is, $P(x_k = 0) = P(x_k = 1) = 0.5$. Therefore, according to the fact that the noise obeys the complex Gaussian distribution with mean value 0 and variance N_0, the prior probability of sending symbol s and receiving r can be obtained by calculating the Euclidean distance $|\mathbf{r} - s|$ in the IQ complex plane,

$$P(\mathbf{r}|s) = \frac{1}{\sqrt{\pi N_0}} e^{-\frac{|\mathbf{r}-s|^2}{N_0}} \tag{11}$$

However, since the symbol in LoRa CSS modulation is a time-related sequence of signals, it can not calculate the LLR of each received bit through the Euclidean distance in the IQ complex plane like M-QAM modulation.

Review of LoRa demodulation in Sec. 2.1, The amplitude spectrum $Z[f]$ in Eq. 3 is the reference for obtaining the demodulation result after de-chirp, each peak in the spectrum is the energy at the different frequency, which encompasses the effects of channel and noise.

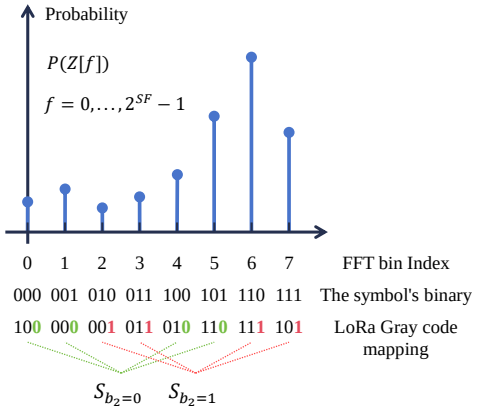


Fig. 6. Example of binary and Gray code mapping for FFT bin when SF=3.

Therefore, we use the spectrum distribution $Z_p[f]$ by demodulating the preamble during detection and alignment to calculate the noise power $N_0 = \frac{1}{2^{SF}-1}\sum_{s\neq 0}|Z_p[s]|^2$.

Then for the demodulation spectrum $Z[f]$ of each chirp in the payload, by calculating the intensity of the signal power and normalizing it to a probability scale, we can map each peak into a probability and obtain an energy-based probability density representation $P(Z[f])$ based spectrum,

$$P(Z[f]) = \frac{e^{|Z[f]|^2-N_0}}{\sum_{s\in[0,2^{SF}-1]} e^{|Z[s]|^2-N_0}}, f = [0, 2^{SF}-1]. \tag{12}$$

That is the form of the prior distribution $P(Z[f])$, where each bin index f is mapped to a Gray symbol containing SF bits. So that, we obtain the $LLR = [LLR_0, ..., LLR_{SF}]$ based on the FFT spectrum of preamble and payload.

$$LLR_k(Z) = \log\frac{\sum_{s\in\mathbb{S}_{b_k=1}} P(Z[s])}{\sum_{s\in\mathbb{S}_{b_k=0}} P(Z[s])}, k = 0, 1, ..., SF \tag{13}$$

where the $\mathbb{S}_{b_k=1}$ is the subset construct by all symbol which the k-th bit is 1, and similarly the $\mathbb{S}_{b_k=0}$.

Due to the Gray code inverse mapping step in the LoRa modulation process, the true codeword corresponding to each FFT bin index should be the Gray symbol mapped after Gray coding, as shown in Fig. 6. After Gray code mapping, we calculate the LLRs for SF bits.

5.2 Confidence Symbol Extraction

After Gray encoding, we perform soft demodulation on the signal of the interleaved block window to obtain an LLR matrix. Based on the Hamming code table (Table 3) used in LoRa, we design a Hamming soft decoding algorithm for soft-output decoding. Algorithm.1 fully leverages the soft information obtained from the demodulation process, mapping the LLR matrix to the probability of correctly decoding different codewords.

After soft Hamming decoding, the Log Likelihood Ratios (LLRs) associated with the data bits are amended based on the decoding outcome and its confidence level. These LLRs essentially represent the reliability of each individual decoded bit. To assess the overall reliability of an entire symbol within an interleaving block (typically containing 4+CR symbols), a symbol-level confidence metric is calculated. This metric is specifically derived by computing the sum of squares of the sf individual bit LLRs. A higher sum-of-squares value indicates greater confidence in the correctness of the entire symbol.

Algorithm 1: Hamming Soft Decoding Algorithm

Input: $CodeTable_{16\times 8}$, $\mathbf{LLR}_{SF\times(4+CR)}$
Output: $DecodeWord$, $Prob$
for $i = 1, \ldots, 16$ **do**
  $cw \leftarrow CodeTable[i] \gg (CR-4)$
  $idx_{same} \leftarrow (cw \wedge (\mathbf{LLR} > 0)) \vee (\neg cw \wedge (\mathbf{LLR} < 0))$
  $idx_{diff} \leftarrow \neg idx_{same}$
  $Prob_{same} \leftarrow \sum_{k\in idx_{same}} |LLR_k|^2$
  $Prob_{diff} \leftarrow \sum_{k\in idx_{diff}} |LLR_k|^2$
  $Prob[i] \leftarrow Prob_{same} - Prob_{diff}$
end
$Prob \leftarrow \max_i Prob$
$DecodeWord \leftarrow CodeTable[i_{max}]$
return $DecodeWord$, $Prob$

Table 3. Hamming Code Table for Different CR

CR	Codeword and Hamming Code			
1	0000|0000=0	0001|1000=24	0010|1000=40	0011|0000=48
	0100|1000=72	0101|0000=80	0110|0000=96	0111|1000=120
	1000|1000=136	1001|0000=144	1010|0000=160	1011|1000=184
	1100|0000=192	1101|1000=216	1110|1000=232	1111|0000=240
2,3,4	0000|0000=0	0001|0111=23	0010|1101=45	0011|1010=58
	0100|1110=78	0101|1001=89	0110|0011=99	0111|0100=116
	1000|1011=139	1001|1100=156	1010|0110=166	1011|0001=177
	1100|0101=197	1101|0010=210	1110|1000=232	1111|1111=255

The confidence symbol selection procedure employs a dynamic thresholding methodology to adapt to fluctuating channel conditions. Initially, we establish an interleaving-block threshold (τ_b) derived from the statistical distribution of symbol confidence metrics within the current processing block, computed as: $\tau_b = \mu_conf + \sigma_conf$. where μ_{conf} represents the mean confidence level and σ_{conf} denotes the standard deviation across all symbols in the processing block. This mechanism facilitates rapid adaptation to immediate signal quality variations within the current block. Subsequently, we implement a global threshold (τ_g) by computing a weighted average of historical interleaving-block thresholds:$\tau_g = \alpha \cdot \tau_g^{(t-1)} + (1-\alpha) \cdot \tau_b^{(t)}$, where $\alpha = 0.8$ represents the weighting coefficient controlling the influence of historical measurements, $\tau_g(t-1)$ denotes the previous global threshold value, and $\tau_{ib}(t)$ indicates the current interleaving-block threshold. This global threshold provides a more stable packet-level confidence assessment by attenuating instantaneous fluctuations, thus enhancing system robustness against ephemeral channel perturbations and noise variations.

Finally, confidence symbols are identified by comparing the calculated confidence level of each symbol against the established global threshold. Symbols whose confidence meets or exceeds this global threshold are selected. These chosen confidence symbols then serve as crucial soft pilots within the iterative demodulation framework. In subsequent iterations, the received signals corresponding to these confidence symbols act as

Algorithm 2: Iterative Soft Demodulate the Block

Input: $iter$, $\hat{h}$, $T = [T_1, \ldots, T_{4+CR}]$
Output: $DecodeWord_{(4+CR)\times SF}$

for $m = 1, \ldots, iter$ **do**
  $LLR_{(4+CR)\times SF} \leftarrow softDemod(T, \hat{h})$
  $LLR_{SF\times(4+CR)} \leftarrow deinterleave(LLR_{(4+CR)\times SF})$
  $DecodeWord_{(4+CR)\times SF}, Prob_{1\times SF} \leftarrow softDecode(LLR, CodeTable)$
  $LLR' \leftarrow amendLLR(LLR, Prob)$
  $symLLR_{1\times(4+CR)} \leftarrow getSymbolLLR(LLR')$
  $Thres_l, Thres_g \leftarrow updateThreshold(symLLR)$
  $ConfdSym \leftarrow getConfidentSymbol(symLLR, Thres_l, Thres_g)$
  if $ConfdSym' = ConfdSym$ **then**
    BREAK
  end
  $\hat{h} \leftarrow channelSense(ConfdSym, T, \hat{h})$
end
return $DecodeWord_{(4+CR)\times SF}$

reference signals used to re-estimate and refine local demodulation parameters (such as LLRs and $\hat{h}$). This iterative process—recalculating LLRs, updating thresholds, selecting confidence symbols, refining parameters—continues until the set of confidence symbols converges (remains unchanged between iterations) or a predefined maximum number of iterations is reached, thereby balancing demodulation accuracy and computational overhead.

Algorithm.2 will output the decoding result $DecodeWord_{(4+CR)\times SF}$ when the confident symbols remain unchanged compared to the previous iteration or when the preset iteration count iter is reached. The default configuration for iteration rounds is set to $iter = 5$, but this can be dynamically adjusted based on the current noise power to balance error rate and computational overhead.

6 SENSING Module

In this section, we use the confidence symbols extracted in the previous Sec. 5 to eliminate the original modulation information in $R(x)$ signal. Furthermore, after removing interference from hardware and the original static vector, we obtain a clean dynamic vector as CSI for sensing.

6.1 Channel State Information Extraction

When the transmitted signal is reflected by stationary objects and sensed objects in the environment, it finally reaches the receiver as a superimposed signal of multipaths signals.

$$Rx(t) = T(t) \cdot e^{j(\theta_c+\theta_s)} \left[H_s(t) + H_d(t)\right] + n(t) \tag{14}$$

θ_c, θ_s are the phase error introduced by Carrier Frequency Offset (CFO) and Sampling Frequency Offset (SFO).

Based on the generated reference signal $Ref(t, \hat{s}) = S(t, \hat{S})$, we can eliminate modulation effects in the time-frequency domain, obtaining channel state measurements that used for sensing, where $R(t)$ is the pilot signals extracted by Sec. 5.2.

$$h_{td} = \frac{R(t)}{ref(t)} = e^{j(\theta_c+\theta_s)} \left[H_s(t) + H_d(t)\right] + n(t) \tag{15}$$

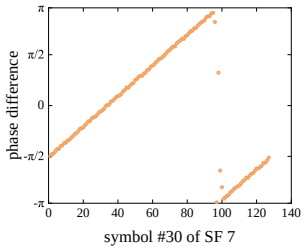


Fig. 7. Phase Difference of Symbol #30 SF=7.

After removing the modulation, the residual signal, represented as Eq. 15, is the channel state measurement between the transmitter and receiver pair. Since H_s corresponds to the static channel effects of the environment, which remain stable within coherent time, we differentiate the above equation to obtain the signal used for sensing.

$$\begin{aligned} Sen(t) &= h_{td}(t+1) - h_{td}(t) \\ &= e^{j(\theta_c+\theta_s)}\left[H_d(t+1) - H_d(t)\right] + n(t) \\ &= e^{j(\theta_c+\theta_s)} H_d(t)(e^{-j2\pi\frac{\Delta d}{\lambda}} - 1) + n(t) \end{aligned} \tag{16}$$

Where Δd represents the change in multipaths distance caused by the movement of the sensing object.

But for fine-grained sensing, the effects of the carrier frequency offset and phase offset must be eliminated.

6.2 Eliminate Offset

The research in[6, 46] shows that the changes in CFO during the package duration did not fluctuate much.

Therefore, we only need to perform a fine-grained CFO estimation after detecting the packet to eliminate CFO during the entire duration of the packet. Specifically, we use the estimation method proposed in[49], using the base up-chirp in Preamble and the base down-chirp in SFD for frame alignment, and then based on the bin index after zero-padding, solve the fine-grained CFO and STO using a system of simultaneous equations.

And since the carrier frequency and sampling frequency are generated by the same oscillator, the sampling frequency offset (SFO) can be eliminated based on the estimated CFO.

However, in addition to these offsets, due to the modulation form of LoRa, the signal frequency will jump to $-BW/2$ when it reaches the highest bandwidth $BW/2$. This situation will occur at the beginning of a new chirp and inside of the modulated chirp.

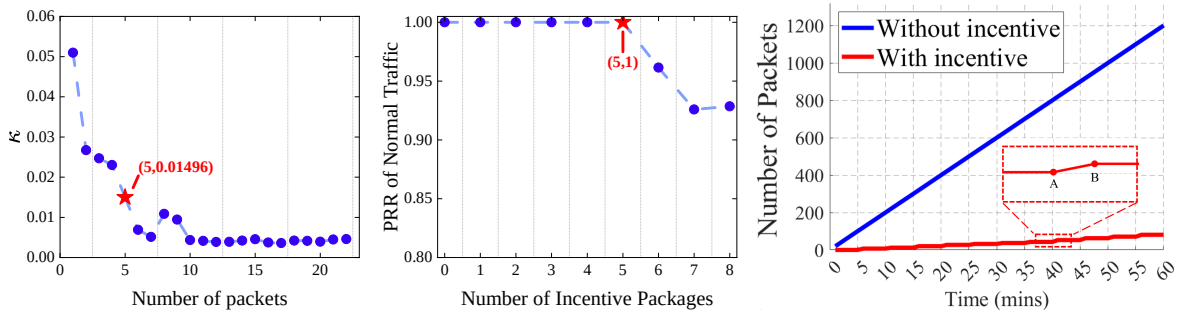
This is actually because the hardware design of LoRa uses a phase-locked loop (PLL)[34] to implement CSS modulation based on phase modulation[45].

Therefore, when the modulator requires switching to new modulation parameters or modifying the phase modulation, the LoRa radio will experience hardware instability, eventually leading to phase jitters as shown in Fig. 7.

Without eliminating the phase jitters caused by this jump, we cannot use continuous IQ samples to perform phase tracking of the sensing object.

Fortunately, the jump moment of a chirp and its symbol s satisfy the relationship $t_{jump} = \frac{2^{SF}-s}{BW}$, So we can again use the symbol obtained based on the demodulation and decode process in Sec. 2.1 to locate their corresponding jump moments. Then use the interpolation method to remove adjacent sample points at the moments. The signal after eliminating the offset is:

$$Sen'(t) = H_d(t)(e^{-j2\pi\frac{\Delta d}{\lambda}} - 1) + n(t) \tag{17}$$



(a) Relationship between Number of Packets and κ

(b) Relationship between Number of Incentive Packets and PRR for Normal Communication

(c) Cumulative Number of Packets in an Hour Before and After Using the κ-based Incentive Strategy

Fig. 8. Analysis of the κ-based Incentive Strategy

7 INCENTIVE Module

In this section, we utilize the sensing entropy mentioned in Sec. 3.3 to evaluate whether packets can complete sensing tasks. Subsequently, we evaluate that additional incentive packets do not significantly impact the throughput under the LoRaWAN protocol, and the energy consumption remains within a controllable range.

In certain complex sensing tasks, the sensing entropy carried by a single packet may not be adequate to meet our requirements. Through experimental studies on respiratory sensing and human walking detection, we discover that these sensing tasks could only be achieved with 90% accuracy when the concentration parameter κ falls below a specific threshold γ.

However, normal communication traffic may not naturally meet this requirement. Therefore, we design an active stimulation mechanism for the SenLoRa system. When insufficient sensing entropy is detected, SenLoRa actively requests nodes to send additional packets until the accumulated sensing entropy reduces the κ value below the threshold γ, ensuring that sensing tasks can be executed with the required accuracy. This mechanism effectively balances communication demands and sensing performance, allowing SenLoRa to flexibly adapt to the requirements of various sensing tasks while guaranteeing communication functionality.

To balance battery overhead, the incentive mechanism is not triggered merely when sensing entropy is deemed insufficient. Instead, the system initially leverages the node's pre-configured periodic uplink transmissions to perform preliminary walking detection by analyzing associated Channel State Information (CSI) fluctuations. The incentive strategy is activated only when CSI variations indicative of human presence are consistently observed across multiple consecutive periodic data packets. This conditional triggering ensures that additional packets are solicited specifically to acquire finer-grained sensing information precisely when a target's presence is strongly suggested.

To determine the threshold γ for sensing entropy, we conduct a respiration sensing experiment, experimental settings are described in Sec. 8.3. After extracting the sensing CSI, we calculated the κ values for sensing using one to many packets. As shown in Fig. 8(a), κ gradually decreases and stabilizes as the number of packets increases, indicating an increase in available sensing information. Based on our calculations, a respiration accuracy of 90% can be achieved with five packets, so we set $\gamma = 0.01496$ for the respiration sensing task. When $\kappa > \gamma$, the gateway sends a beacon to prompt the node to transmit 5 consecutive packets. The value of γ may vary for different scenarios and sensing tasks.

We quantify the impact of the incentive strategy on normal communication by measuring the relationship between the number of incentive packets and the PRR of normal communication. In our experiment, we configure a total of 11 nodes, all using the LoRaWAN LMAC protocol[13]. Ten of these nodes are used for normal communication, with each node sending two packets per minute. The remaining node serves as an incentive node, sending 0 to 8 packets per minute. Data is collected over one minute, and the PRR results for normal communication are shown in Fig. 8(b). It is observed that the PRR is 100% without incentive packets, and decrease to 92.85% when 8 incentive packets are sent. This result indicates that an increase of incentive packets has little effect on the PRR of normal communication. When $\kappa = \gamma$, 5 incentive packets are needed, which does not influence normal LoRa communication traffic.

Fig. 8(c) investigates the impact of incentive strategies on energy consumption. We conducted real-world data collection for breath sensing under conditions with and without the incentive mechanism. In our experiment, we monitored human breathing patterns over one-hour periods under two distinct conditions.The control group operated without incentives, continuously transmitting 30 packets per minute for breath sensing. In contrast, the experimental group implemented our proposed incentive strategy, initially sending only one packet per minute to detect human walking. Once two consecutive packets confirmed human walking, the incentive mechanism was activated, resulting in the cumulative transmission of 5-10 sensing packets (as shown at points A and B).The results demonstrate that implementing the incentive strategy significantly reduced packet transmission, with approximately 93% fewer packets sent compared to the non-incentive approach. This substantial reduction in data transmission directly translates to lower energy consumption while maintaining effective sensing capabilities.

8 EVALUATION

In this section, we will present the real-world implementation of SenLoRa and evaluate its performance in both communication and sensing aspects.

8.1 Implementation

The LoRa node consists of an STM32 NUCLEO-L476RG development board with an embedded Semtech SX1261 module, while a USRP SDR B210 serves as the receiver to implement SenLoRa. The receiver is responsible for collecting LoRa signal samples. The captured signal files are transmitted to a laptop via an Ethernet cable. Signal processing algorithms are implemented in C++ and GNU Radio, with subsequent post-processing done in MATLAB. Unless otherwise specified, the following context assumes this configuration.

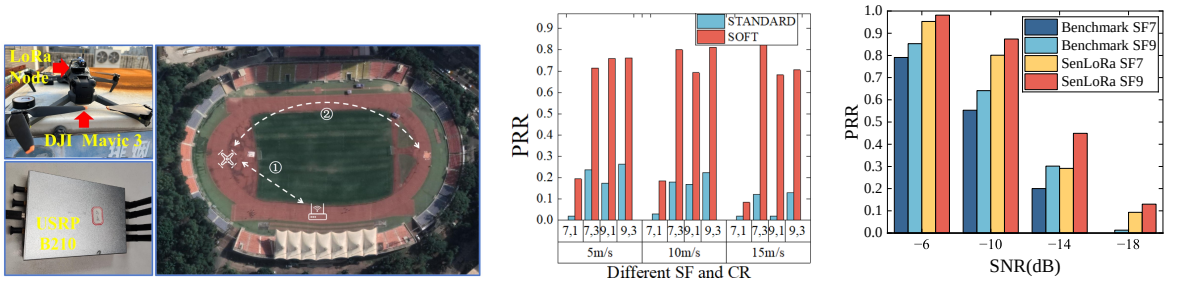
8.2 Communication Performance

We conduct an Unmanned Aerial Vehicle (UAV) communication experiment in extreme environments to compare the performance of soft and standard demodulation.

Experiment Setup: We use the SX1261 node for transmission with a transmit power setting of 5, while employing the USRP B210 as the receiver with a receiving sensitivity set to 50. A lower power setting is configured to evaluate performance under baseline conditions. As shown in Fig. 9 (a), the nodes are attached to the drone, which is then flown at different speeds over a distance of approximately 200 meters, continuously transmitting about 30 packets during the flight. The demodulation approach follows standard open-source methodology[49].

We also explore the effectiveness of iterative soft demodulation as discussed in Sec. 5.2. We perform 5 rounds of iterative demodulation on the data with velocities V = 5, 10, 15 m/s, SF = 7, 9, and CR = 1, 3. An iteration count of 0 corresponds to hard demodulation. We calculate packet reception rate $PRR = \frac{\text{No. of correctly decoded packets}}{\text{Total packets}}$ and Bit Error Rate $BER = \frac{\text{No. of incorrect bits}}{\text{Total bits}}$.

Result: As shown in Fig. 9 (b, c), the PRR of soft demodulation are significantly better than standard demodulation method in all tested configurations. Moreover, SenLoRa exhibits the capability to provide reliable



(a) DJI UAV and flight trajectory

(b) PRR for different SF/CR at different speeds

(c) PRR performance under different SNR and SF

Fig. 9. Experimental setup and communication performance

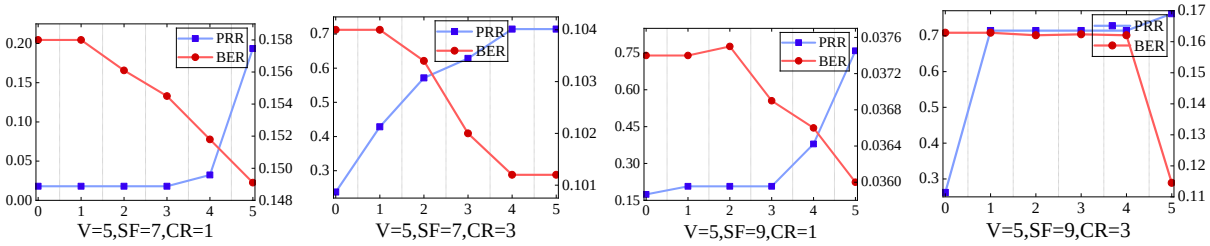


Fig. 10. Performance of iterative soft demodulation for a UAV moving at a velocity of 5 m/s, under different Spreading Factors (SF) and Coding Rates (CR). The horizontal axis shows the number of iterations, with 'iter=0' representing standard hard demodulation. The results demonstrate that as the number of iterations increases, the Packet Reception Rate (PRR) improves while the Bit Error Rate (BER) decreases, validating the effectiveness of the iterative approach.

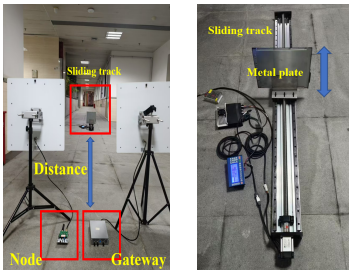
long-range communication in a variety of IoT applications, even under varying SNR and SF conditions. Fig. 10 demonstrates the iterative performance improvement of SenLoRa. As the number of iterations increases, the Packet Reception Rate (PRR) gradually rises while the Bit Error Rate (BER) decreases. These opposing trends are reasonable, as PRR represents the proportion of packets that pass verification, while BER is defined as the proportion of error bits. This illustrates the effectiveness of iterative soft demodulation.

8.3 Sensing Performance

In this section, we evaluate the sensing performance of SenLoRa by testing a metal sheet (35 cm × 35 cm) on a sliding rail under varying distances, sliding ranges, and obstacle conditions to validate the effectiveness of the CSI-based sensing strategy. The results demonstrate the accuracy of SenLoRa in perception tasks.

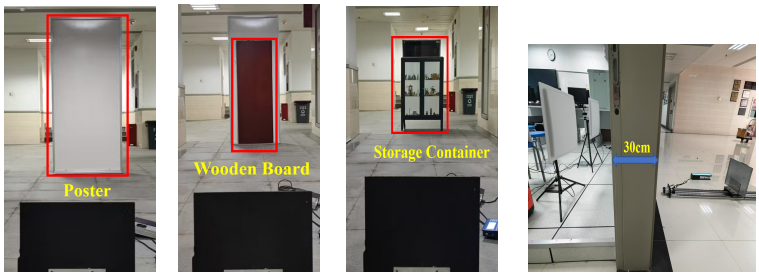
Experiment Setup: We evaluated the sensing capability of SenLoRa in an indoor environment. As illustrated in Fig. 11(a), the node, gateway, and sliding rail were arranged as shown, with Fig. 11(b) providing a detailed view of the sliding rail, where the sliding distances were set to 1 cm and 10 cm, respectively. To simulate potential obstructions in cold storage environments, we employed three different test cases: a poster (Fig. 12(a)), a poster combined with a wooden board (Fig. 12(b)), and a storage container (Fig. 12(c)). Furthermore, Fig. 12(d) shows the system performance in through-wall sensing scenarios.

Metric: The ground-truth frequency of the sliding rail motion is calculated by dividing the sliding speed by the distance, with the unit in Beats Per Minute (BPM). The measured value is obtained through envelope



(a) (b)

Fig. 11. Gateway placement and slider details

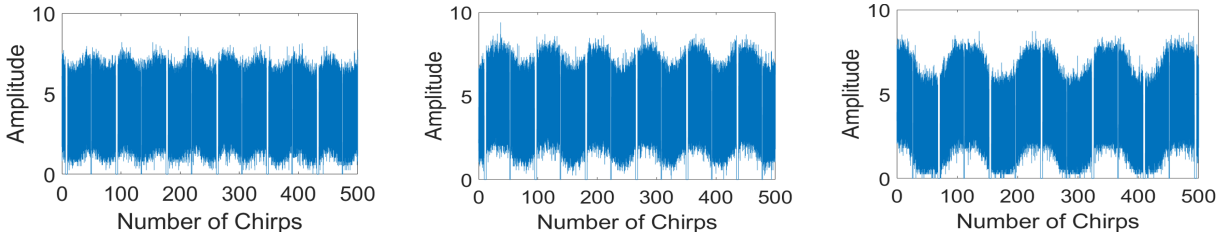


(a) (b) (c) (d)

Fig. 12. Various testing scenarios

downsampling followed by Fast Fourier Transform (FFT). Unless otherwise specified, each data point represents the mean absolute error (MAE) of three measurements

The signal ratio is obtained by dividing dual-antenna signals at the receiver, enhancing LoRa sensing capability by eliminating environmental interference. SenLoRa uses a single antenna to calculate CSI.



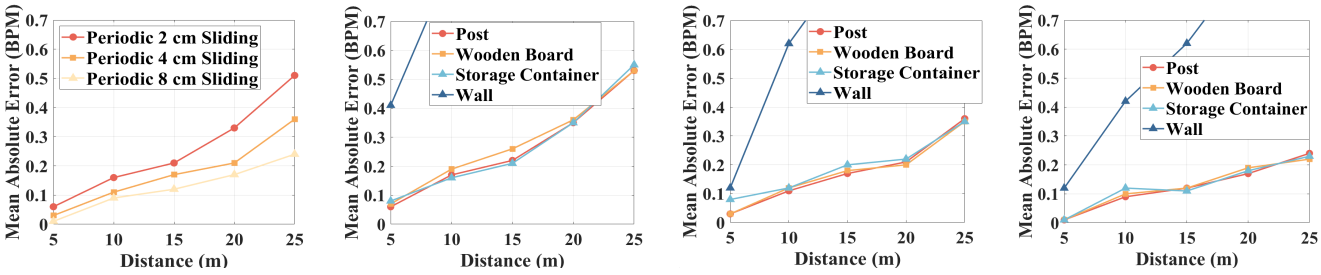
(a) 2 cm (b) 4 cm (c) 8 cm

Fig. 13. At a fixed distance of 15 meters, we characterize how the signal envelope evolves with increasing chirp counts for periodic sliding movements across varying travel distances (2 cm/4 cm/8 cm)

Effectiveness of CSI Sensing: Fig. 13 demonstrates that an 8 cm sliding distance induces more pronounced and smoother envelope waveform variations compared to a 4 and 2 cm. This also explains why larger sliding distances are more easily detectable at longer ranges, as observed in Fig. 14(a). As shown in Fig. 14(b-d), benefiting from LoRa's inherent characteristics, we find that common obstacles such as posters, wooden boards, and counter have minimal impact on sensing performance. Even when encountering a 30 cm thick solid wall, an 8 cm sliding motion can still achieve detectable sensing at a 10 m distance (we consider the sensing capability lost when the mean absolute error exceeds 0.4 BPM). When the sliding distance reaches 50 cm, we can distinctly detect its periodic variations even at a 50-meter range. For conciseness, further elaboration is omitted here.

8.4 Case Study

We evaluated the performance of SenLoRa by conducting human walking detection and respiration sensing in real-world scenarios. We invited 10 volunteers aged 20-25 to participates in the following two experiments: human respiration sensing and human walking detection. All experiments involving human participants were conducted with ethical oversight.



(a) Different sliding distances

(b)2 cm silding under interference

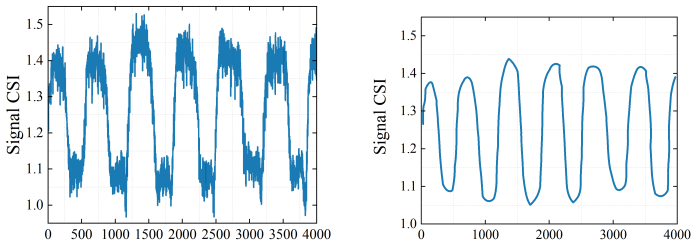
(c) 4 cm silding under interference

(d) 8 cm silding under interference

Fig. 14. Subfigure (a) illustrates the sensing range and capability of SenLoRa at different sliding distances, while subfigures (b)-(d) demonstrate their performance under interference.



Fig. 15. Warehouse scenario



(a) SenLoRa

(b) Dual-antenna method

Fig. 16. Comparison of methods

Experiment Setup: Fig. 15 show a real warehouse environment. We used the HKH-11C respiratory belt as the ground truth for respiration monitoring. To validate SenLoRa's capability in detecting moving humans, we collected both respiratory data and movement data under three walking patterns. During data acquisition, no moving objects were present near the receiver. For each data collection instance, we acquire continuous signal samples with a 20-second duration, which provides sufficient temporal resolution to capture both transient events and periodic patterns in the monitored activities. This sampling duration was empirically determined to achieve optimal balance between computational resource utilization and sensing accuracy.

Metric: We use the following metrics to evaluate the performance of SenLoRa. In human walking detection, (i) True Positive Rate (TPR) is defined as the correct detection rate of the samples collected when a human is present, i.e., $TPR = \frac{TP}{(TP+FN)}$, where TP is the number of positively detected true positive samples and FN is the number of positive samples falsely detected as negative; (ii) True Negative Rate (TNR) is defined as the correct detection ratio of samples collected when no human is present, i.e., $TNR = \frac{TN}{(TN+FP)}$, where TN is the number of negative samples that are correctly detected, FP is the number of negative samples that are falsely detected as positive. In human respiration detection, the ground truth of respiration rate is calculated as $\frac{number\ of\ breaths}{recorded\ time}$, with the unit as beats per minute (BPM).

Overall performance. Fig. 16 reproduces the methodology of Zhang et al[53]., which utilizes dual-antenna simultaneous reception to eliminate instantaneous noise and bias in CSI acquisition, thereby obtaining purer CSI measurements than SenLoRa. However, this improvement shows minimal effect on both walking detection and respiration frequency monitoring. Fig. 17 demonstrates a practical sensing scenario where the gateway

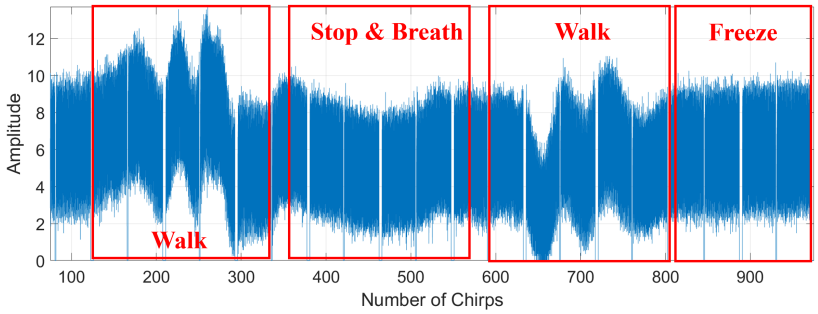
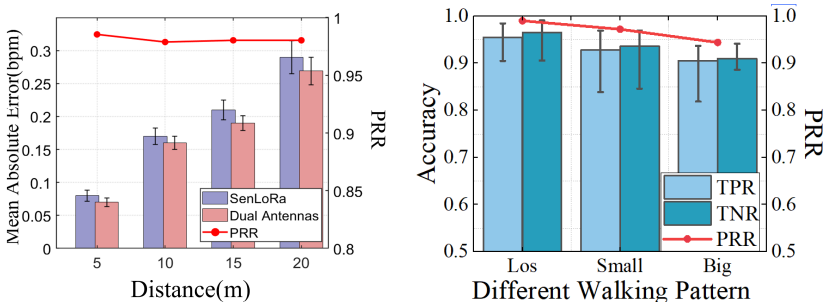


Fig. 17. This presents the CSI waveform evolution during sequential activity transitions: (i) ambulatory motion (walking), (ii) stationary breathing, (iii) resumed ambulation, and (iv) motionless non-breathing state. This tripartite differentiation demonstrates decimeter-scale motion resolution while maintaining spectral separation between locomotor and cardiopulmonary signatures.



(a) The Error of respiration rate versus distance to the transceivers, the red line represents PRR of SenLoRa

(b) Values of TPR and TNR for different walking patterns, the red line represents PRR of SenLoRa

Fig. 18. Experiment results of human respiration and walking detection on different conditions.

is located 10 m away from the subject. The signal envelope clearly reflects the following sequence of human activities: walking to a complete stop, maintaining stationary breathing for several seconds, resuming walking, and finally stopping without breathing. The distinct waveform patterns accurately correspond to each physical activity phase, effectively validating SenLoRa's capability in temporal activity resolution and motion-respiration differentiation.

Effectiveness of range: To evaluate the influence of distance between human target and transceivers on sensing performance, we adjust the distance 4 times within the same environment, designated as 5 m, 10 m , 15 m and 20 m, respectively. Fig. 18(a) illustrates the BPM results, the mean BPM errors of SenLoRa at 4 distances are 0.08, 0.17 , 0.21 and 0.29, respectively, achieving similar precision as previous work[53], while PRR of SenLoRa are higher than 94%. It can be observed that SenLoRa performs well at different distances.

Effectiveness of different walking patterns: In real-world environments, humans may walk in different patterns. Now we explore the impact of walking patterns on presence detection. We compare the results of SenLoRa with the ground truth, with TPR and TNR under the three patterns shown in Fig. 18(b), where the values of TPR are 95.48%, 92.8% and 90.56%, respectively, and the values of TNR are 96.57%, 93.61% and 91%, and PRR of SenLoRa are above 94%. The high accuracy of SenLoRa suggests its capability for classification tasks, affirming

the effectiveness of using CSI for sensing. The results indicate little impact of varying human walking patterns on sensing accuracy.

8.5 Summary

To help readers quickly grasp SenLoRa's takeaways, we have created table 4 outlining the main specifications:

Table 4. Performance summary of the SenLoRa system

Metric / Aspect	Key Result / Value
Communication Enhancement	
Packet Reception Rate (PRR)	Improvement up to 4× vs. Standard LoRa
PRR during Sensing	> 94% maintained
Wireless Sensing	
Respiration Rate Error (MAE)	As low as 0.2 bpm (15 m)
walking Detection Accuracy	> 90% (TPR & TNR)
Energy Efficiency (Incentive)	
Sensing Packet Overhead	~ 93% reduction vs. continuous transmission
Network Throughput Impact	< 8% PRR drop for 8 incentive pkts/min
Node Energy Cost	No extra cost beyond incentive packet TX

9 RELATED WORK

LoRa Communication: LoRa as a popular technology in LPWAN, has been widely applied in communication between IoT devices, this has also stimulated many studies based on the LoRa physical layer, such as[49] presents a comprehensive understanding of the LoRa Physical Layer Protocol (PHY) and reveals the fundamental reasons for the performance gap, and proposes the first full-stack LoRa PHY implementation with a provable performance guarantee. [45] analyzes the problem of LoRa phase jitter and used it as a feature to implement a low-power demodulation method at the sub-sampling level. [44] proposes a lightweight node authentication scheme named SLoRa for LoRa networks by leveraging two physical layer features in Carrier Frequency Offset(CFO) and spatial-temporal link signature. [32] verifies the implementation and effectiveness of four latest works on LoRa packet collision, and realized a unified platform to evaluate these efforts.

[8] presents the LoRaFFEC protocol, which proposes fragmentation and Forward Error Correction to the LoRaWAN standard.

With LoRaFFEC, any payload length can benefit from a better Data Delivery Rate (DDR) for a wide range of channel-level packet losses. And[50] also realize the LDPC in LoRa using the Soft Belief Propagation(SBP) for decoding. [35] solves the problem of packet collision, it is further proposed that through joint decoding, more codewords can be corrected through combination.

LoRa Sensing: LoRa technology has been widely used in various sensing scenarios. Existing LoRa-based research falls into two categories: LoRa sensing capability enhancement and LoRa sensing tasks implementation. By adopting technologies such as signal ratio, the sensing range of LoRa has been extended from the original tens of meters to hundreds of meters. Zhang et al.[53] realize a 25 m respiration sensing range using signal ratio, and further design a LoRa beamforming technique to support multi-target sensing[54]. In sensing scenarios, LoRa is usually used to perform tasks that can provide quantitative feedback, such as localization[2, 5, 17, 18, 27, 29, 31], tracking[9] and soil moisture sensing[4]. Ren et al.[31] achieve high-precision localization by calculating LoRa channel state information (CSI) and using it as the input to the super-resolution algorithm ESPIRT after calibration.

Zhang et al.[4] realize wide-area soil moisture sensing by extracting the dielectric permittivity of soil from LoRa phase readings.

Existing methods often require dual-antennas or continuous packet transmission, which are incompatible with commercial LoRa nodes and inefficient in spectrum usage.

ISAC: Integrated Sensing And Communication: Existing ISAC systems can be categorized based on deployment methods and communication-sensing fusion approaches. Generally, the sensing side is widely deployed in a transceiver model to achieve fusion through echoes. SenLoRa chooses Rx-to-Tx sensing at the receiver and realizes sensing based on time-domain CSI. For fusion strategies, unlike WiFi which inserts pilots, we choose to reuse communication waveforms to avoid additional operations.

LoRa can realize long-range sensing[47]. However, most current communication-sensing fusion methods involve incorporating communication capabilities to sensing systems[7, 30]. For example, Zhang et al.[23] achieved two-way LoRa communication by carrying LoRa signals in FMCW radar waveforms, retaining the radar's sensing resolution. Wang et al.[38] uses dual-antennas and a help node to track transmitter movement while maintaining communication capability, but this approach is impractical in real-world scenarios. Therefore, loading sensing functions onto communication can more fully and effectively utilize existing infrastructure. Methods that load communication functionalities onto sensing systems, however, often struggle to achieve usable communication performance. Through analysis, the sensing tasks loaded onto LoRa by SenLoRa can cover most existing application scenarios.

10 Discussion

Multi-target Sensing. Multi-target sensing remains a fundamental challenge in contactless wireless perception systems, with three primary approaches currently available. The first method employs Independent Component Analysis (ICA)[51], which requires prior knowledge of target quantity and must satisfy three conditions: (i) multi-antenna receiver arrays, (ii) linear signal combination across antennas, and (iii) more observed signal sources than targets. The second approach utilizes multi-dimensional domain analysis (spatial/temporal/spectral)[48] for WiFi-based target separation, relaxing some ICA constraints while still requiring multiple antennas. The third method[21] necessitates each user to carry their own WiFi-enabled device (e.g., smartphone). However, since commercial LoRa nodes typically feature only a single antenna, none of these methods may be directly applicable. Consequently, our future work will focus on developing CSI separation techniques based on the distinct signal characteristics of different targets to overcome this limitation in LoRa-based sensing systems.

Power Consumptions. The power efficiency of SenLoRa is optimized through two key design strategies: First, we deliberately avoid any hardware or software modifications to the transmitter nodes, thereby maintaining their original power consumption profile. The computationally intensive tasks are exclusively offloaded to the LoRa gateways, which benefit from continuous mains power supply and can readily accommodate the additional processing overhead. Second, our carefully designed incentive mechanism prevents nodes from maintaining persistent packet transmission, significantly extending their operational lifespan by minimizing unnecessary energy expenditure.

11 CONCLUSION

In this paper, we analyze the contradiction between communication and sensing and propose SCR. We then introduce SenLoRa, a system that enables sensing using LoRaWAN uplink frames. SenLoRa innovatively reuses communication facilities and data packets for sensing. Our design incorporates an iterative soft demodulation algorithm, which aims to precisely extract modulation information and acquire reference signals, thereby mitigating the modulation impact on received signals. Additionally, we provide incentive strategies to facilitate

the completion of other tasks. Real-world experimental results demonstrate that SenLoRa not only enhances decoding performance but also supports a wide range of sensing applications.

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12 APPENDIX

Here we present the detailed derivation procedure for the sample entropy calculation, often used to quantify the regularity and predictability inherent in a time series $H_d(t) = \{H_d(1), H_d(2), ..., H_d(N)\}$. Sample entropy assesses the complexity by estimating the negative natural logarithm of the conditional probability that two sequences similar for m points remain similar for $m+1$ points, within a tolerance r, crucially excluding self-matches.

First, the time series is decomposed into overlapping template vectors (subsequences) of length m and $m+1$. The m-length vectors are defined as:

$$X_m(i) = [H_d(i), H_d(i+1), ..., H_d(i+m-1)] \tag{18}$$

for $i = 1, ..., N-m+1$. There are $N-m+1$ such vectors. Similarly, the $(m+1)$-length vectors are:

$$X_{m+1}(i) = [H_d(i), H_d(i+1), ..., H_d(i+m)] \tag{19}$$

for $i = 1, ..., N-m$. There are $N-m$ such vectors.

Next, we define a distance metric to compare these vectors. The Chebyshev distance (maximum absolute difference between corresponding elements) is commonly used:

$$d[X_k(i), X_k(j)] = \max_{0 \le p \le k-1} |H_d(i+p) - H_d(j+p)| \tag{20}$$

where k can be m or $m+1$. Two vectors $X_k(i)$ and $X_k(j)$ are considered similar if their distance is less than a predefined tolerance r:

$$d[X_k(i), X_k(j)] < r \tag{21}$$

We then count the number of similar pairs, ensuring self-matches are excluded ($i \neq j$). Let $B^m(r)$ be the total number of pairs of m-length vectors $(X_m(i), X_m(j))$ that satisfy the similarity condition $d[X_m(i), X_m(j)] < r$, where $1 \le i, j \le N-m+1$ and $i \neq j$. Let $A^{m+1}(r)$ be the total number of pairs of $(m+1)$-length vectors $(X_{m+1}(i), X_{m+1}(j))$ that satisfy the similarity condition $d[X_{m+1}(i), X_{m+1}(j)] < r$, where $1 \le i, j \le N-m$ and $i \neq j$.

The ratio $\frac{A^{m+1}(r)}{B^m(r)}$ provides an estimate of the conditional probability that if two sequences are similar for m points, they will also remain similar when the next point (up to $m+1$ points) is included.

Finally, the Sample Entropy (SE) is defined as the negative natural logarithm of this conditional probability estimate:

$$SE(H_d(t), m, r) = -\ln\left(\frac{A^{m+1}(r)}{B^m(r)}\right) \tag{22}$$

Where m is the embedding dimension and r is the tolerance. $B^m(r)$ represents the count of similar pattern pairs of length m (excluding self-matches), and $A^{m+1}(r)$ represents the count of similar pattern pairs of length $m+1$ (excluding self-matches). A lower SE value indicates higher regularity and predictability in the time series, while a higher value suggests greater complexity or randomness.

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